رفدي — Sequence Diagrams

Internal Reference Document — All 9 Sequences

For internal team use only | Version 1.0

Overview

This document is the complete internal reference for all system interaction sequences in the رفدي platform. It includes all 9 key use cases showing how React, FastAPI, MySQL, and external services interact.

#	Sequence	Description
1	**Login**	Shows login flow for both Admin (validated against
2	**Create Booking + Payment**	The most critical sequence
3	**Register Company**	New company registers on the platform
4	**Browse & Filter Warehouses**	Renter searches and filters available warehouses
5	**Admin: Suspend Company**	Platform Admin suspends a company account
6	**Owner: Add Warehouse**	Warehouse Owner adds a new warehouse listing including image upload
7	**Owner: Toggle Warehouse Active/Inactive**	Owner toggles warehouse visibility in search results using IsActive flag
8	**Send Notification (Email/SMS)**	Internal notification flow triggered by booking or payment events
9	**Dual-Role Switch (Owner ↔ Renter)**	Company with both roles switches between Owner and Renter views

Sequence 1 — Login

Shows login flow for both Admin (validated against .env) and Company (validated against MySQL). Admin is hard-coded in .env and does not exist in the database.

#	Actor → Receiver	Action
1	User → React	Submits email and password via login form
2	React → FastAPI	POST /auth/login with credentials
3a	FastAPI → .env	[Admin] Compare email+password against .env values
3b	FastAPI → MySQL	[Company] SELECT user WHERE email, verify password hash
4	FastAPI → React	Return JWT token with role (admin / owner / renter)
5	React → User	Store JWT, redirect to role-specific dashboard
ERR	FastAPI → React	[Fail] Return 401 Unauthorized

Figure 3: Login Sequence Diagram

Sequence 2 — Create Booking + Payment

The most critical sequence. Covers date overlap validation, commission calculation (5% renter / 7% owner), Moyasar payment processing, and notification dispatch.

#	Actor → Receiver	Action
1	Renter → React	Selects warehouse, enters start and end dates
2	React → FastAPI	POST /bookings with warehouse ID and dates
3	FastAPI → MySQL	checkOverlap() — query existing confirmed bookings
ERR	FastAPI → React	[Overlap] Return 409 — dates unavailable
4	FastAPI	[No overlap] calculateCommission() — 5% renter, 7% owner
5	FastAPI → MySQL	INSERT booking record with commission fields
6	FastAPI → React	Return bookingID, redirect to payment screen
7	React → FastAPI	POST /payments with card details
8	FastAPI → Moyasar	processPayment() — call Moyasar API
9	Moyasar → FastAPI	Return payment confirmed
10	FastAPI → MySQL	UPDATE payment status to Paid
11	FastAPI	send() — trigger Email/SMS notification to Owner and Renter
12	FastAPI → React	Return booking confirmed, show success screen

Figure 4: Create Booking + Payment Sequence Diagram

Sequence 3 — Register Company

New company registers on the platform. No admin approval required — company can login immediately after registration.

#	Actor → Receiver	Action
1	Company → React	Fills registration form: name, commercial registration, email, password, roles
2	React → FastAPI	POST /auth/register with form data
3	FastAPI	Validate fields: no empty fields, valid email format, no negative values
4	FastAPI → MySQL	Check duplicate email — SELECT WHERE email
ERR	FastAPI → React	[Duplicate] Return 400 — email already exists
5	FastAPI	[New] Hash password using bcrypt
6	FastAPI → MySQL	INSERT company record + INSERT roles (owner / renter / both)
7	MySQL → FastAPI	Return companyID
8	FastAPI → React	Return 201 Created
9	React → Company	Redirect to login page — company can login immediately

Figure 5: Register Company Sequence Diagram

Sequence 4 — Browse & Filter Warehouses

Renter searches and filters available warehouses. Only IsActive=1 warehouses are returned.

#	Actor → Receiver	Action
1	Renter → React	Enters search keyword and selects filters (location, size)
2	React → FastAPI	GET /warehouses?location=&size= with JWT header
3	FastAPI	Validate JWT token, extract user role
4	FastAPI → MySQL	SELECT WHERE IsActive=1 AND filters
5	MySQL	Apply filters, return matching warehouses

#	Actor → Receiver	Action
6	MySQL → FastAPI	Return warehouse list
7	FastAPI → React	Return JSON warehouses[]
8	React → Renter	Render warehouse cards with details

Figure 6: Browse & Filter Warehouses Sequence Diagram

Sequence 5 — Admin: Suspend Company

Platform Admin suspends a company account. FastAPI verifies admin JWT before allowing the action.

#	Actor → Receiver	Action
1	Admin → React	Clicks Suspend button on a company row
2	React → FastAPI	PATCH /companies/{id}/suspend with JWT
3	FastAPI	Verify JWT role=admin — reject if not admin
4	FastAPI → MySQL	UPDATE companies SET status=Suspended WHERE id
5	MySQL → FastAPI	OK
6	FastAPI → Email/SMS	send() — suspension notice to company
7	Email/SMS → FastAPI	Delivered
8	FastAPI → React	Return 200 OK
9	React → Admin	Show Suspended badge on company row

Figure 7: Admin: Suspend Company Sequence Diagram

Sequence 6 — Owner: Add Warehouse

Warehouse Owner adds a new warehouse listing including image upload.

#	Actor → Receiver	Action
1	Owner → React	Fills Add Warehouse form with all required fields
2	React → FastAPI	POST /warehouses (multipart/form-data) with JWT

#	Actor → Receiver	Action
3	FastAPI	Validate fields: no negative values, required fields present
4	FastAPI	Save image to disk/cloud, generate image URL
5	FastAPI → MySQL	INSERT warehouse with all fields including imageURL
6	MySQL → FastAPI	Return warehouseID
7	FastAPI → React	Return 201 Created
8	React → Owner	Show new warehouse in Manage Warehouses table

Figure 8: Owner: Add Warehouse Sequence Diagram

Sequence 7 — Owner: Toggle Warehouse Active/Inactive

Owner toggles warehouse visibility in search results using IsActive flag.

#	Actor → Receiver	Action
1	Owner → React	Clicks toggle switch on warehouse row
2	React → FastAPI	PATCH /warehouses/{id}/toggle with JWT
3	FastAPI	Verify JWT, confirm requester owns this warehouse
4	FastAPI → MySQL	UPDATE warehouses SET IsActive = !IsActive WHERE id
5	MySQL → FastAPI	Return new IsActive value
6	FastAPI → React	Return 200 OK + new status
7	React → Owner	Update toggle UI to reflect new state

Figure 9: Owner: Toggle Warehouse Active/Inactive Sequence Diagram

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Sequence 8 — Send Notification (Email/SMS)

Internal notification flow triggered by booking or payment events. Saves to MySQL and sends via external Email/SMS provider.

#	Actor → Receiver	Action
1	FastAPI	Trigger event detected (booking created / payment confirmed)

#	Actor → Receiver	Action
2	FastAPI → MySQL	INSERT notification record (userID, type, message, IsRead=false)
3	MySQL → FastAPI	Return notificationID
4	FastAPI → Email/SMS	send(type, message, userID) via provider API
5	Email/SMS → FastAPI	Delivered confirmation
6	React → FastAPI	[Later] User opens app — GET /notifications
7	FastAPI → MySQL	SELECT notifications WHERE userID ORDER BY CreatedAt DESC
8	MySQL → FastAPI	Return notifications[]
9	FastAPI → React	Return JSON with unread count
10	React → User	Show bell badge with unread count + dropdown list

Figure 10: Send Notification (Email/SMS) Sequence Diagram

Sequence 9 — Dual-Role Switch (Owner ↔ Renter)

Company with both roles switches between Owner and Renter views. Role switching is handled client-side using JWT claims.

#	Actor → Receiver	Action
1	Company → React	Clicks role switcher in Navbar
2	React	Read roles[] array from JWT payload (no API call needed)
3	React	Switch activeRole state in React Context
4	React → FastAPI	[If new data needed] GET /warehouses or GET /bookings with new role context
5	FastAPI → MySQL	Query data relevant to new role
6	MySQL → FastAPI	Return role-specific data
7	FastAPI → React	Return JSON response
8	React → Company	Render new role dashboard with appropriate navigation and data

Figure 11: Dual-Role Switch (Owner ↔ Renter) Sequence Diagram